🧪 Lab 2 : Secure & Scalable VLAN Design (Router-on-a-Stick)

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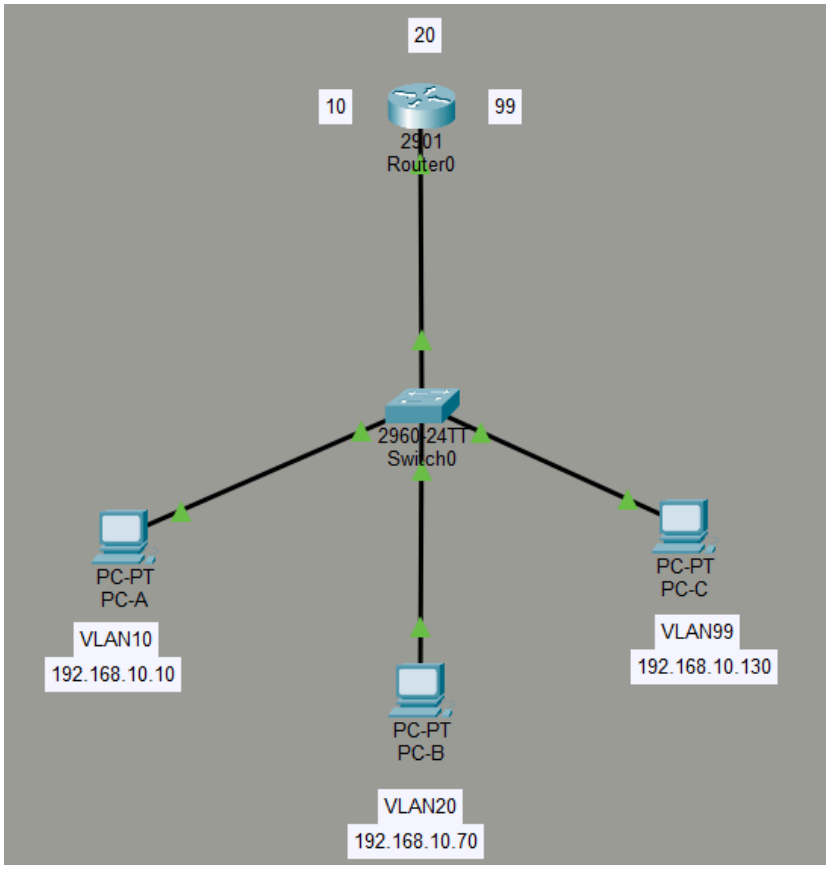
📊 TOPOLOGY DIAGRAM



📋 ADDRESSING TABLE

Network Summary

- **Major Network:** 192.168.10.0/24
- **Subnet Mask (/26):** 255.255.255.192
- **Subnets Used:**
  - VLAN 10 → 192.168.10.0/26
  - VLAN 20 → 192.168.10.64/26
  - VLAN 99 → 192.168.10.128/26

Device Addressing

Device	Interface	VLAN	IP Address	Subnet Mask	Default Gateway
R1	G0/0.10	10	192.168.10.1	255.255.255.192	N/A
R1	G0/0.20	20	192.168.10.65	255.255.255.192	N/A
R1	G0/0.99	99	192.168.10.129	255.255.255.192	N/A
S1	VLAN 99	99	192.168.10.131	255.255.255.192	192.168.10.129
PC-A	NIC	10	192.168.10.10	255.255.255.192	192.168.10.1
PC-B	NIC	20	192.168.10.70	255.255.255.192	192.168.10.65
PC-C	NIC	99	192.168.10.130	255.255.255.192	192.168.10.129

🔧 LAB INSTRUCTION STEPS

PART 1: SWITCH CONFIGURATION

Step 1: Basic Switch Setup

```
S1>enable
S1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
S1(config)#hostname S1
S1(config)#
```

Step 2: Create VLANs

```
S1(config)#vlan 10
S1(config-vlan)#name USERS
S1(config-vlan)#exit
S1(config)#
S1(config)#vlan 20
S1(config-vlan)#name SERVERS
S1(config-vlan)#exit
S1(config)#
S1(config)#vlan 99
S1(config-vlan)#name MANAGEMENT
S1(config-vlan)#exit
S1(config)#
S1(config)#vlan 999
S1(config-vlan)#name BLACKHOLE
S1(config-vlan)#exit
S1(config)#
```

```
S1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig0/2
10   USERS                            active    Fa0/1
20   SERVERS                          active    Fa0/2
99   MANAGEMENT                       active    Fa0/3
999  BLACKHOLE                        active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                               Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                               Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                               Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                               Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                               Fa0/24
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

Step 3: Secure Unused Ports (Security First)

```
S1(config-if-range)#shutdown

%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/9, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/11, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/12, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/13, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/14, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/15, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/16, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/17, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/18, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/19, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/20, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/21, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/22, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/23, changed state to administratively down
```

Step 4: Assign Access Ports

```
S1(config)#
S1(config)#interface fa0/1
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 10
S1(config-if)# no shutdown
S1(config-if)#exit
S1(config)#
S1(config)#interface fa0/2
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 20
S1(config-if)# no shutdown
S1(config-if)#exit
S1(config)#
S1(config)#interface fa0/3
S1(config-if)# switchport mode access
S1(config-if)# switchport access vlan 99
S1(config-if)# no shutdown
S1(config-if)#exit
S1(config)#
S1(config)#
S1(config)#interface range fa0/4 - 24
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 999
S1(config-if-range)# shutdown
S1(config-if-range)#exit
```

```
S1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------
1    default                          active    Gig0/2
10   USERS                            active    Fa0/1
20   SERVERS                          active    Fa0/2
99   MANAGEMENT                       active    Fa0/3
```

Step 5: Configure Trunk to Router

```
S1(config)#interface g0/1
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk allowed vlan 10,20,99
S1(config-if)# no shutdown
S1(config-if)#exit
S1(config)#do show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      10,20,99

Port        Vlans allowed and active in management domain
Gig0/1      10,20,99

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      99
```

Step 6: Configure Management SVI

```
S1(config)#interface vlan 99
S1(config-if)# ip address 192.168.10.131 255.255.255.192
S1(config-if)# no shutdown
S1(config-if)#exit
S1(config)#
S1(config)#ip default-gateway 192.168.10.129
S1(config)#end
S1#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
S1#
S1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual up                    up
FastEthernet0/3        unassigned      YES manual up                    up
FastEthernet0/4        unassigned      YES manual administratively down down
FastEthernet0/5        unassigned      YES manual administratively down down
FastEthernet0/6        unassigned      YES manual administratively down down
FastEthernet0/7        unassigned      YES manual administratively down down
FastEthernet0/8        unassigned      YES manual administratively down down
FastEthernet0/9        unassigned      YES manual administratively down down
FastEthernet0/10       unassigned      YES manual administratively down down
FastEthernet0/11       unassigned      YES manual administratively down down
FastEthernet0/12       unassigned      YES manual administratively down down
FastEthernet0/13       unassigned      YES manual administratively down down
FastEthernet0/14       unassigned      YES manual administratively down down
FastEthernet0/15       unassigned      YES manual administratively down down
FastEthernet0/16       unassigned      YES manual administratively down down
FastEthernet0/17       unassigned      YES manual administratively down down
FastEthernet0/18       unassigned      YES manual administratively down down
FastEthernet0/19       unassigned      YES manual administratively down down
FastEthernet0/20       unassigned      YES manual administratively down down
FastEthernet0/21       unassigned      YES manual administratively down down
FastEthernet0/22       unassigned      YES manual administratively down down
FastEthernet0/23       unassigned      YES manual administratively down down
FastEthernet0/24       unassigned      YES manual administratively down down
GigabitEthernet0/1     unassigned      YES manual up                    up
GigabitEthernet0/2     unassigned      YES manual down                  down
Vlan1                  unassigned      YES manual administratively down down
Vlan99                 192.168.10.131  YES manual up                    up
```

PART 2: ROUTER CONFIGURATION

Step 7: Basic Router Setup

```
R1>enable
R1#
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
R1(config)#hostname R1
R1(config)#
```

Step 8: Enable Physical Interface

```
R1(config)#interface g0/0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
```

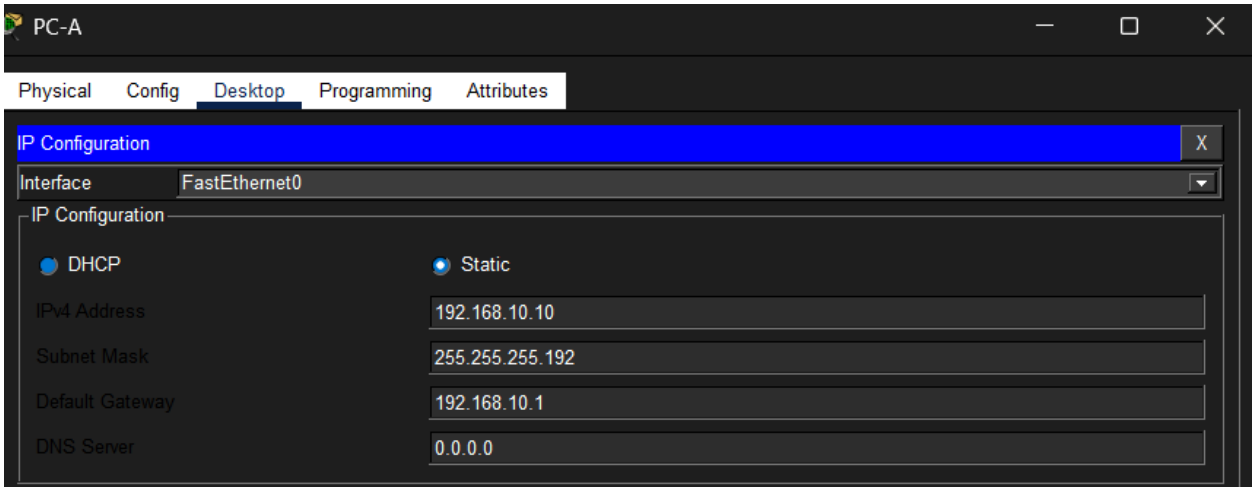
Step 9: Configure Router-on-a-Stick Subinterfaces

```
R1(config)#interface g0/0.10
R1(config-subif)#encapsulation dot1Q 10
R1(config-subif)#ip address 192.168.10.1 255.255.255.192
R1(config-subif)#exit
R1(config)#
R1(config)#interface g0/0.20
R1(config-subif)#encapsulation dot1Q 20
R1(config-subif)#ip address 192.168.10.65 255.255.255.192
R1(config-subif)#exit
R1(config)#
R1(config)#interface g0/0.99
R1(config-subif)#encapsulation dot1Q 99
R1(config-subif)#ip address 192.168.10.129 255.255.255.192
R1(config-subif)#exit
R1(config)#


R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#
R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R1#
```

PART 3: END DEVICE CONFIGURATION

ตั้งค่า IP, Subnet Mask, Default Gw ของแต่ละ PC

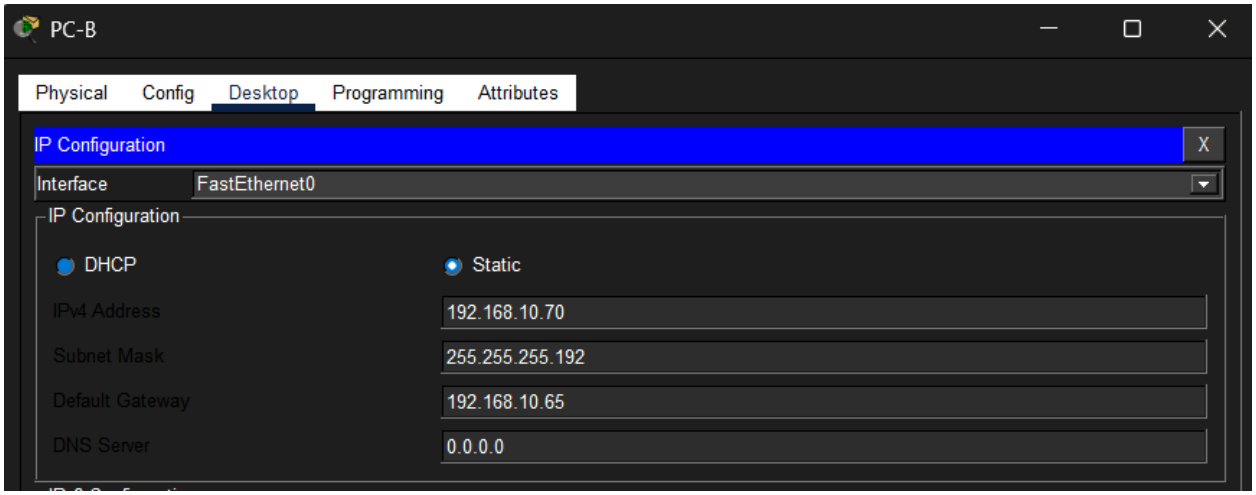
PC-A



ตรวจสอบค่าต่างๆบน cmd ด้วยคำสั่ง ipconfig

```
FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::2E0:F9FF:FE61:C91D
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.10.10
   Subnet Mask.....................: 255.255.255.192
   Default Gateway.................: ::
                                     192.168.10.1
```

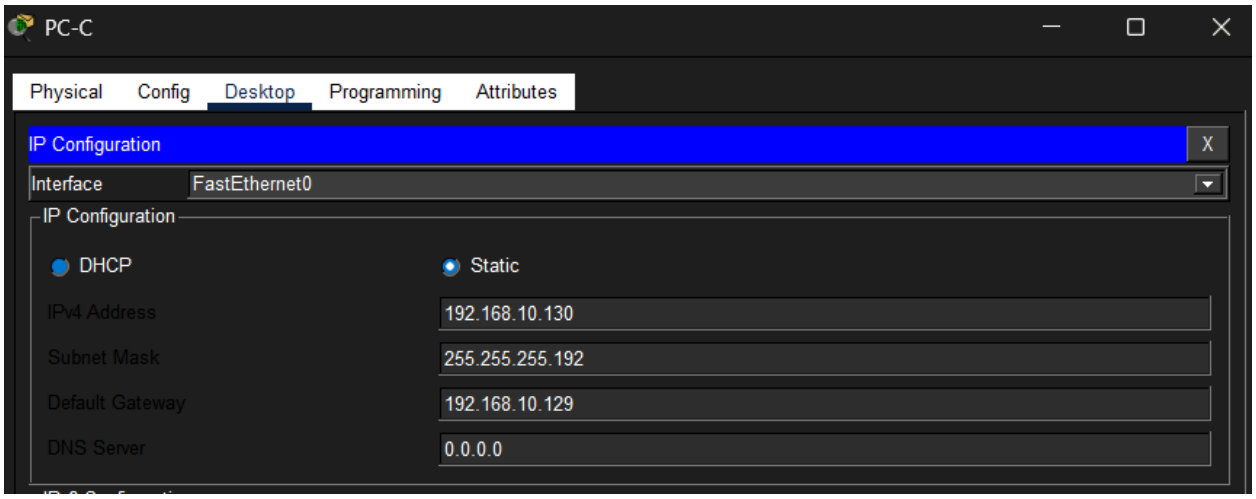
PC-B



ตรวจสอบค่าต่างๆบน cmd ด้วยคำสั่ง ipconfig

```
FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::260:5CFF:FEC4:D692
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.10.70
   Subnet Mask.....................: 255.255.255.192
   Default Gateway.................: ::
                                     192.168.10.65

Bluetooth Connection:
```

PC-C



ตรวจสอบค่าต่างๆบน cmd ด้วยคำสั่ง ipconfig

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::290:2BFF:FEB8:7674
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.10.130
   Subnet Mask.....................: 255.255.255.192
   Default Gateway.................: ::
                                     192.168.10.129
```

PART 4: CONNECTIVITY VERIFICATION

ทดสอบการรับส่งข้อมูลระหว่าง End device แต่ละเครื่อง

Part 4.1) From PC-A

Step 1) ping 192.168.10.1 (PC-A)

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Step 2) ping 192.168.10.70 (PC-B)

```
C:\>ping 192.168.10.70

Pinging 192.168.10.70 with 32 bytes of data:

Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.70:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Step 3) ping 192.168.10.130 (PC-C)

```
C:\>ping 192.168.10.130

Pinging 192.168.10.130 with 32 bytes of data:

Reply from 192.168.10.130: bytes=32 time<1ms TTL=127
Reply from 192.168.10.130: bytes=32 time<1ms TTL=127
Reply from 192.168.10.130: bytes=32 time<1ms TTL=127
Reply from 192.168.10.130: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Part 4.2) From PC-B

Step 1) ping 192.168.10.65 (PC-B)

```
C:\>ping 192.168.10.65

Pinging 192.168.10.65 with 32 bytes of data:

Reply from 192.168.10.65: bytes=32 time<1ms TTL=255
Reply from 192.168.10.65: bytes=32 time<1ms TTL=255
Reply from 192.168.10.65: bytes=32 time<1ms TTL=255
Reply from 192.168.10.65: bytes=32 time=3ms TTL=255

Ping statistics for 192.168.10.65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 0ms
```

Step 2) ping 192.168.10.10 (PC-A)

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time=3ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 0ms
```

Step 3) ping 192.168.10.130 (PC-B)

```
C:\>ping 192.168.10.130

Pinging 192.168.10.130 with 32 bytes of data:

Reply from 192.168.10.130: bytes=32 time<1ms TTL=127
Reply from 192.168.10.130: bytes=32 time<1ms TTL=127
Reply from 192.168.10.130: bytes=32 time<1ms TTL=127
Reply from 192.168.10.130: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Part 4.3) From PC-C

Step 1) ping 192.168.10.129 (PC-C)

```
C:\>ping 192.168.10.129

Pinging 192.168.10.129 with 32 bytes of data:

Reply from 192.168.10.129: bytes=32 time<1ms TTL=255
Reply from 192.168.10.129: bytes=32 time<1ms TTL=255
Reply from 192.168.10.129: bytes=32 time<1ms TTL=255
Reply from 192.168.10.129: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Step 2) ping 192.168.10.10 (PC-A)

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Step 3) ping 192.168.10.70 (PC-B)

```
C:\>ping 192.168.10.70

Pinging 192.168.10.70 with 32 bytes of data:

Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127
Reply from 192.168.10.70: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.70:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Part 4.4) From S0

```
S1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig0/2
10   USERS                            active    Fa0/1
20   SERVERS                          active    Fa0/2
99   MANAGEMENT                       active    Fa0/3
999  BLACKHOLE                        active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
S1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      10,20,99

Port        Vlans allowed and active in management domain
Gig0/1      10,20,99

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      10,20,99
```

Part 4.4) From R0

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.10.65   YES manual up                    up
GigabitEthernet0/0.99  192.168.10.129  YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

PART 5: TROUBLESHOOTING CHALLENGE

Scenario A: Wrong VLAN Assignment

- Move PC-B to the wrong VLAN

```
S1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig0/2
10   USERS                            active    Fa0/1
20   SERVERS                          active
99   MANAGEMENT                       active    Fa0/2, Fa0/3
999  BLACKHOLE                        active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

- Test connectivity

```
C:\>ping 192.168.10.65

Pinging 192.168.10.65 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.65:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

- Identify failure point

  Scenario A แสดงให้เห็นว่า หาก PC ถูก assign ไป VLAN ผิด
  จะไม่สามารถสื่อสารกับ gateway และอุปกรณ์ใน subnet เดิมได้
  แม้ IP address จะถูกต้องก็ตาม

- Restore correct VLAN

```
C:\>ping 192.168.10.65

Pinging 192.168.10.65 with 32 bytes of data:

Reply from 192.168.10.65: bytes=32 time<1ms TTL=255
Reply from 192.168.10.65: bytes=32 time<1ms TTL=255
Reply from 192.168.10.65: bytes=32 time<1ms TTL=255
Reply from 192.168.10.65: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Scenario B: Missing VLAN on Trunk

- Remove VLAN 99 from trunk

```
S1>enable
S1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      10,20,99

Port        Vlans allowed and active in management domain
Gig0/1      10,20,99

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      10,20,99

S1#Vlans allowed on trunk
         ^
% Invalid input detected at '^' marker.

S1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
S1(config)#interface g0/1
S1(config-if)#switchport trunk allowed vlan 10,20
S1(config-if)#end
S1#
%SYS-5-CONFIG_I: Configured from console by console
show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      10,20

Port        Vlans allowed and active in management domain
Gig0/1      10,20

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      10,20
```

- Test access to S1

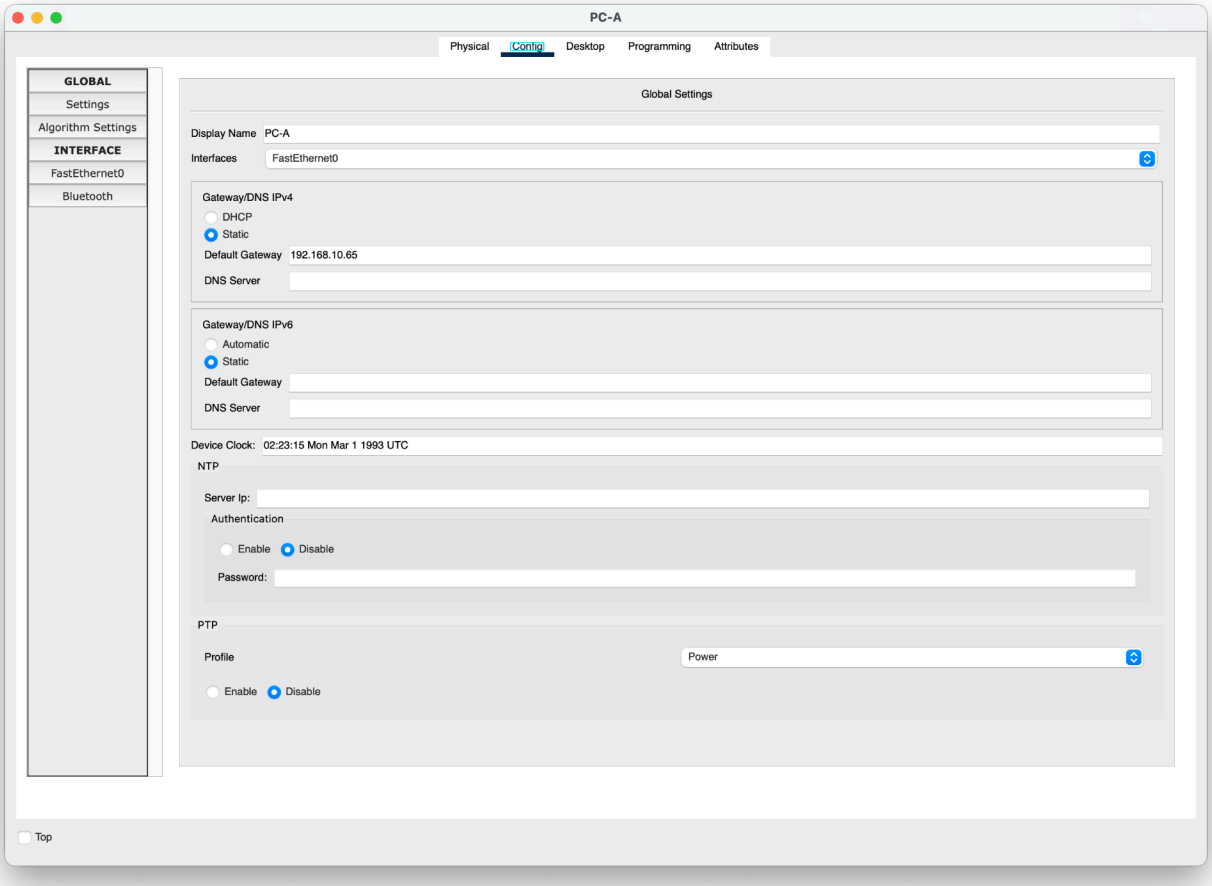
```
C:\>ping 192.168.10.129

Pinging 192.168.10.129 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.129:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

- Explain why management fails
  - VLAN 99 ไม่ถูก allow บน trunk
  - Management traffic ถูก drop
  - Router ↔ Switch สื่อสารกันไม่ได้
- Restore configuration

```
S1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
S1(config)#interface g0/1
S1(config-if)#switchport trunk allowed vlan 10,20,99
S1(config-if)#end
S1#
%SYS-5-CONFIG_I: Configured from console by console
show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      10,20,99

Port        Vlans allowed and active in management domain
Gig0/1      10,20,99

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      99
```

Scenario C: Wrong Gateway

- Change PC-A gateway



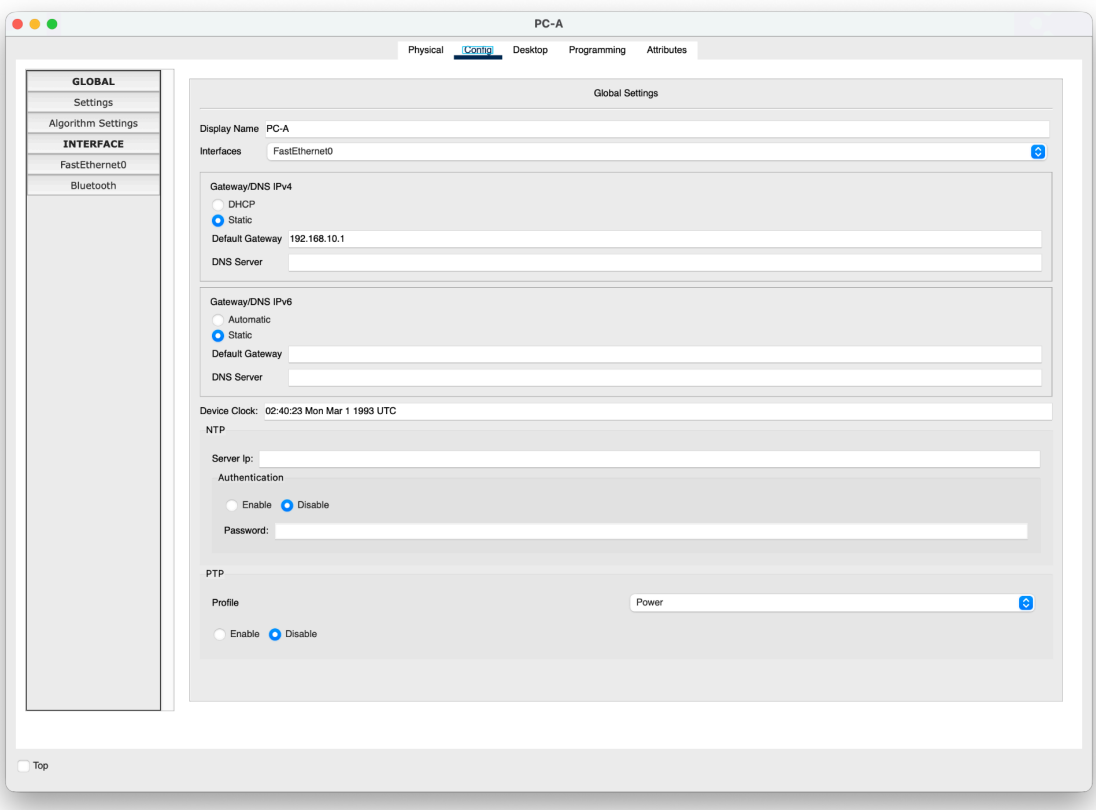
- Observe ping failure

```
C:\>ping 192.168.10.70

Pinging 192.168.10.70 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.70:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

- Use `ipconfig` and `ping`

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::2E0:F9FF:FE61:C91D
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.10.10
   Subnet Mask.....................: 255.255.255.192
   Default Gateway.................: ::
                                     192.168.10.254

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0

C:\>
C:\>ping 192.168.10.70

Pinging 192.168.10.70 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.
```

- Fix gateway



📝 LAB DELIVERABLES

Worksheet 1: Configuration Check

Checkpoint	Command	Expected	Actual	Pass/Fail
VLANs exist	show vlan brief	VLANs 10,20,99	พบ VLAN 10 (USERS), VLAN 20 (SERVERS), VLAN 99 (MANAGEMENT) แสดงสถานะ active	✅ Pass
Trunk active	show int trunk	VLANs allowed	Port Gi0/1 trunking (802.1Q), allowed VLANs 10,20,99	✅ Pass
Router subifs	show ip int br	up/up	G0/0.10 (192.168.10.1), G0/0.20 (192.168.10.65), G0/0.99 (192.168.10.129) status up/up	✅ Pass
Inter-VLAN ping	ping	Success	ทุก PC สามารถ ping ข้าม VLAN ได้ครบ (PC-A, PC-B, PC-C)	✅ Pass

---

Worksheet 2: Design Understanding

Answer briefly:

1. **Why is VLAN 99 used for management?**
   -ใช้ VLAN 99 สำหรับ management เพื่อแยก traffic การบริหารอุปกรณ์ (เช่น Switch) ออกจาก traffic ผู้ใช้งานทั่วไป เพิ่มความปลอดภัยและง่ายต่อการจัดการระบบ

2. **Why is VLAN 999 assigned to unused ports?**
   -VLAN 999 ใช้สำหรับพอร์ตที่ไม่ได้ใช้งานเพื่อป้องกันการเชื่อมต่อที่ไม่ได้รับอนุญาต หากมีอุปกรณ์เสียบเข้ามาจะไม่สามารถสื่อสารกับ VLAN อื่นได้ (security best practice)

3. **What does 802.1Q tagging do?**
   -802.1Q tagging ใช้ติด tag VLAN ID ไปกับ Ethernet frame เพื่อให้หลาย VLAN สามารถส่งผ่านลิงก์ trunk เดียวกันได้ และให้ปลายทางรู้ว่า frame นั้นอยู่ VLAN ใด

4. **Why does each VLAN require a different subnet?**
   -แต่ละ VLAN เป็นคนละ broadcast domain จึงต้องใช้ subnet ที่แตกต่างกัน เพื่อให้ Router สามารถทำ inter-VLAN routing ได้อย่างถูกต้องและลด broadcast traffic

---

🔍 VERIFICATION COMMANDS CHECKLIST

Router

- show run ✅
- show ip route ✅
- show arp ✅

Switch

- show vlan brief ✅
- show interfaces trunk ✅
- show mac address-table ✅

PCs

- Ipconfig ✅
- arp -a ✅
- Ping ✅

🎓 ASSESSMENT QUESTIONS

Which device performs inter-VLAN routing?

<u>ตอบ</u> **Router**เป็นอุปกรณ์ที่ทำให้ VLAN ต่าง ๆ คุยกันได้ เพราะแต่ละ VLAN คือคนละเครือข่าย การจะส่งข้อมูลข้ามเครือข่ายต้องผ่านอุปกรณ์ที่มีความสามารถ “Routing”

Why can PCs in different VLANs communicate?

<u>ตอบ</u> เพราะมี **Inter-VLAN Routing** ทำหน้าที่รับข้อมูลจาก VLAN หนึ่ง แล้วส่งไปยังอีก VLAN หนึ่ง เหมือนมี “ล่ามแปลภาษา” ทำให้เครือข่ายที่แยกกันอยู่ สามารถสื่อสารกันได้

What breaks if the trunk is removed?

<u>ตอบ</u> **VLAN** จะไม่สามารถส่งข้อมูลผ่านสวิตช์ระหว่างอุปกรณ์/อาคาร/ชั้นได้
เพราะลิงก์แบบ trunk คือ “เส้นทางหลัก” ที่รวมหลาย VLAN ไว้ในสายเดียว ถ้าเอาออก VLAN ต่าง ๆ จะถูกตัดขาดทันที

Why is VLAN 1 avoided?

<u>ตอบ</u> เพราะ **VLAN 1** ถูกตั้งเป็นค่ามาตรฐานของทุกอุปกรณ์ ทำให้เสี่ยงด้านความปลอดภัย เช่น ใครเสียบสายมาก็อาจเข้าเครือข่ายได้ทันที จึงนิยมแยกไปใช้ VLAN อื่นเพื่อลดความเสี่ยง

What security risk exists if unused ports remain active?

<u>ตอบ</u> ใครก็สามารถเสียบสายเข้าใช้งานเครือข่ายได้เลย เช่น แขก หรือคนไม่หวังดี เสียบสายบนพอร์ตว่างที่ยังเปิดอยู่ แล้วเข้าถึงระบบภายใน ทำให้เสี่ยงถูกแฮ็กหรือดึงข้อมูล